



BIN381 PROJECT MILESTONE 2 - DATA PREPROCESSING REPORT

Socioeconomic Development & Household Well-being in South Africa (SDHSA)



Group 5

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Github Repository: <https://github.com/Henri-Claassen/BIN381>

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1. Introduction

1.1 The decision problem

The project examines how digital access can improve education and help people get better jobs in South Africa. The stakeholder is the National Treasury, which must decide how to prioritise limited budgets between digital access subsidies and continued education spending. The central question is whether the two are complements (they strengthen each other) or substitutes (one doesn't affect the other). (Oyedemi and Choung, 2020; Statistics South Africa, 2025a)

Milestone 1 revealed different areas of the data that needs to be tested:

Area	Question	Measured by
A	How Digital access relates to Education	Household internet and computer access (D08) and education level, attendance and institution type (D02)
B	How Education relates to Employment	Education level (D02) and employment status (D03)
C	How Digital access and Education relate to Employment	Interaction between education and digital access in the employment model
D	How income is affected by digital access	Reported salary among employed people (D03) (Exploratory)

2. Data Selection and Analytical Grain

2.1 Datasets selected

Four of the eleven supplied datasets are used. A dataset is included only if it measures an area of the research pathway and can be linked to the same people at the chosen grain.

Dataset	Decision	Reason
D01 Population Profile	Included	Demographic controls (age, sex, population group, province, settlement type, metro), general information about the population.
D02 Learning Profile	Included	Education level, current attendance, institution type, needed to compare education with employment status
D03 Economic Participation	Included	Employment status and salary needed to answer questions about salary level.
D08 Connectivity Access	Included	The only dataset measuring household internet, computer and smartphone access, needed to determine digital access.
D04, D05, D06, D07, D10	Excluded	Household-level datasets that do not measure an area of the pathway (housing, water, energy, food security, transport).
D09 Household Resources	Excluded	Its only digital column duplicates D08, and its income columns partly reveal the employment outcome.
D11 Labour Market	Excluded	A separate survey. It shares 0 of D01's 20,940 household IDs and not a single PSU, and its grain is person-per-quarter. It cannot be linked at any level

2.2 Analytical grain

One row = one person aged 15–64, identified by household_id and person_number with their household's digital-access values attached. (Statistics South Africa, 2025b)

Both outcomes, education and employment, belong to individuals, so a person-level grain is required: a person's education must be compared with that same person's employment status. Digital access is measured per household and is attached to every person in that household.

2.3 Record selection

Criterion	Persons	Reason
All persons (after removing exact duplicates)	70,440	This is to have clean records, which can be split as shown below.
Under 15	–19,803	Below working age, employment status cannot be applied.
65 and older	–5,224	Above working age, retirement so employment stats cannot be applied.
Aged 15–64 (retained)	45,413	Statistics South Africa's working-age definition, t covers 19,780 households

2.4 Attribute selection and weighting

A column is kept only if it measures an area, is needed to link, or is needed to interpret a link descriptively. Of the 83 source columns, 29 were kept after column selection, and some of the reasons for dropping 58 columns include no information (survey_year is always 2024), duplication of another dataset, unusable values, disclosure risk (metro_code), or leakage.

3. Cleaning and Quality Treatment

3.1 Identifier precision: the cross-cutting issue

Problem: household_id is an 18-digit number, but R stores numbers as doubles, which are exact only to about 15–16 digits. This means values are rounded which end up giving duplicate rows, which is an issue. (R Core Team, 2026)

How it was handled: every household_id and person_number column is read as text.

Before and After: when it was read as numbers D08 had 20,684 distinct households which goes up to 20,940 when read as text, all 18 digits are now preserved and every dataset returns 20,940 distinct households.

3.2 Completeness

Problem	Treatment	Effect and verification
25 rows in D01 have a missing household_id	Each row was matched to one of the 25 people who appear in D02/D03 but not in D01, using person_number, sex, province and the person weight.	All 25 IDs recovered, each candidate used exactly once. The PSU equals the first 11 digits of the recovered ID and the age agrees with D03. So no rows were added or removed.
90 missing and 18 invalid ages in D01 (-7, -3, 150, 175)	Replaced from D03, after confirming D02 and D03 agree on every person and contain no invalid ages, and that D01 agrees with them wherever it is valid.	Age filtered to between 0 and 110 with no missing values. No person was deleted.
120 missing education_level_code in D02	Filled from D03, which has the same variable for the same people.	D02 and D03 never disagree and D03 has no gaps, so education level is complete.
60 missing mobile_internet_code in D08	Left missing, there is no dictionary code to assign and no second source to repair them with.	Documented in the data dictionary; affects 139 of 45,413 person rows

3.3 Uniqueness and duplication

Problem: Milestone 1 found rows sharing a person key in D01 (23), D02 (44) and D03 (40). Genuine duplicate records had to be separated from legitimate repeated identifiers.

Treatment and effect: every D01 “duplicate” turned out to be a row with a missing household_id: blank identifiers grouped different people together, and the problem disappeared once the identifiers were recovered. In D02 and D03, every key duplicate was an exact copy of another row, with no conflicting values, so distinct() removed 22 rows from D02 and 20 from D03.

Verification: after treatment all three person level datasets now contain exactly the same 70,440 people, each key appears once, and no exact duplicates remain. Repeated household_id values remain in the person files, which is correct: one household contains several people.

3.4 Validity and special values

Special values were classified from evidence before anything was changed, because treating a “do not know” code as an ordinary category would distort results, while deleting those rows would discard the same people's valid answers. (World Health Organization, 2023)

Value	Classification	Evidence and treatment
education_level_code 98 (No schooling) and 28 (Other)	Valid categories	Real education levels, they were kept as categories.
education_level_code 29, education_attendance_code 3, computer_asset_code 9	No response or missing	“Do not know” / “Unspecified” are not categories, so they receive no label.
education_level_code 99, fixed/mobile_internet_code 9	Undocumented or missing	Not in the data dictionary, so no meaning can be assigned.
institution_type_code 88, reason_not_attending_code 88	“Not applicable” level	The question was not asked of this person.
education_attendance_code “N/A”, “Unknown”, “not recorded” (35 rows)	Text placeholders or missing	None appear in the dictionary. “N/A” cannot be read “not applicable”, since “not applicable” occurs only for children aged 0–4, while the placeholders occur at ages up to 72.
lab_salary_raw 888888888 and 999999999 (52,129 rows)	Placeholders they are missing, with a reason column.	A cross-tabulation against employment status shows 888888888 acts as “not applicable” for people without a job and 999999999 as non-response among the employed. A new salary column holds real values only, and salary_status records why a value is absent.

The salary treatment corrects Milestone 1's conclusion that 74% of salary data was unusable: that figure counted people who cannot have a salary at all. Of the working age, 18,036 of 18,711 (96%) report a usable salary.

3.5 Outliers and suspicious observations

Only the three genuine measurements (age, household_smartphones and salary) were reviewed for outliers, using histograms and boxplots. Statistical outliers were treated as candidates only, and kept unless domain evidence showed the value was impossible. (World Health Organization, 2023)

Variable	Finding and treatment
Age	No outliers remain after the fixing the completeness issues now there are no points outside the boxplot fences.
Household_smartphones	15 households report a negative count, which is impossible. These were set to missing and are traceable. The household is kept because its other connectivity answers are valid. The negative values could not be repaired by removing the minus sign, because several would then exceed the dictionary maximum of 13. High counts were checked against household size, no household has more smartphones than members, so they are possible and were kept
Salary	Strongly right-skewed. High values up to R1,400,000 are genuine high earners and were kept. Of the low outliers, only the 19 values under R100 cannot plausibly be a monthly salary. These were flagged rather than deleted, because the pay period is not documented in the source.

3.6 Recoding and data types

Every retained `_code` column received a label column of the same name without the suffix, built with `factor()` using the data dictionary's labels. Codes classified as non-response are left without a label, so they become missing in the label column while the original code remains visible. The `_code` columns are never modified, so every label can be traced back to its source value.

4. Integration and Construction

4.1 Keys and cardinality

The four datasets were cleaned at their own grain **before** any join, so that duplicates could not multiply rows. Each join states its expected cardinality in dplyr's relationship argument, which stops the pipeline if a key matches more rows than expected, and the person-level joins use unmatched = "error", which stops the pipeline if any row would be dropped.

Join	Key	Cardinality	Rows after	Unmatched
D01 and D02 and D03	household_id + person_number	One-to-one	45,413	0
D08 is also joined	household_id	Many-to-one	45,413	1,160 households with no working-age member (expected)

Verification: the row count stays at 45,413 through all three joins, every person matches a D02, D03 and D08 record, no duplicate columns appear, and every household's members share a single D08 record.

The 1,160 unmatched D08 households are households whose members are all under 15 or over 64. They are outside the analysis population by design, not a key problem, and remain available in D08_clean.csv for household-level description.

4.2 Why D11 was not integrated by another route

D11 measures employment and education for working-age people, so alternatives to a row-level join were considered. Aggregating D11 to province and attaching it to each person would add a value fully determined by province_code, which D01 already contains, so it adds no information at the person level. D11 also contains no digital-access columns, so a parallel analysis could only repeat the education-to-employment link without testing the moderating role of connectivity. The employment outcome therefore comes from a single source (D03), which is recorded as a limitation.

4.3 Engineered features

Twenty-four features were constructed. The features can be built before the train/test split without leakage. The percentages below are weighted with person_weight and are descriptive justification only.

Feature	Construction	Why it adds value
education_band	Groups education_level_code into 7 bands (No schooling, Primary, Some secondary, Matric, Post-school certificate/diploma, Degree, Other/unknown)	Employment rises sharply from matric upward (49.5%, 67.9%, 82.1% employed) while the three bands below matric are (28–32%)
matric_plus	1 for matric or any post-school qualification, 0 below matric, missing for unrankable codes	Gives a single education term for the interaction test and one interpretable odds ratio
any_home_internet	1 if the household has fixed or mobile internet, 0 if both are "No"	Tells us whether a household has internet or not.

Feature	Construction	Why it adds value
digital_access_score	Counts fixed internet + mobile internet + computer + at least one smartphone (0–4)	Captures the depth of access that a single yes/no hides: 31.9% employed at 0 rising to 66.6% at 4
edu_digital_group	Combines matric_plus and any_home_internet into four groups plus “Education unknown”	Helps show how education and internet access relates to employment for tree models and Power BI.
smartphone_access	household_smartphones divided by household_size, grouped as None, fewer than 1 per member and 1 or more per member	Helps determine if having more smartphones in a household affect employment or not.
settlement_type	Combines geo_type and metro into Metro urban, Non-metro urban, Traditional, Farms	Removes overlap between two location columns and merges groups that are too small.
household_size, n_children_under15, n_elderly_65plus, n_working_age, dependency_ratio, elderly_in_household	Counted from D01 before the age filter removed children and older people	How having children responsibilities affect employment, will having more children lead to less employment?
female_x_children	female × n_children_under15	How different genders are affected after having children. Employment falls from 62.7% to 34.3% for women but only 67.0% to 50.4% for men.
other_member_matric	Whether another working-age household member has matric or higher	Does it affect employment rate if more than 1 person is educated in a household.
age_centred, age_centred_squared	age – 40, and its square	Employment is not linear in age. (Employment does not just keep going up and up with age, it peaks between 20-30 and falls off after 45-54, we normalise this so model can predict more accurately.
log_salary	Natural log of salary for employed people with a reported salary of at least R100	Salary is strongly right-skewed (mean about twice the median). On the log scale the distribution is roughly bell-shaped, so a few high earners cannot dominate the income model.

4.4 Data dictionary

The complete data dictionary is in “Cleaned Datasets Data Dictionary.md”. It documents every retained source variable and every engineered variable together with the columns excluded from modelling and the reasons. It is kept in the repository so that it can be updated with the code.

5. Modelling Preparation and Leakage Control

5.1 Target definition and class balance

The modelling target is employed: Employed versus Unemployed and it is restricted to the labour force. People who are not economically active are excluded because they are largely students and people close to retirement who are not looking for work, so the digital job-search mechanism in the business problem does not apply to them. (Statistics South Africa, 2025b)

Of the 28,846 people in the labour force, 104 were dropped because their household has a missing digital-access answer, leaving 28,742. Where 18,641 are employed and 10,101 are unemployed. This is a mild imbalance of roughly 2:1, which `glm()`, `rpart()` and `randomForest()` can fit directly, so no resampling (for example SMOTE or down-sampling) needs to be applied.

5.2 Identifiers, leakage and unsuitable predictors

Columns were removed from the modelling dataset for four reasons:

- **Leakage:** the salary columns exist only for employed people (a reported salary is 100% employed), and `reason_not_attending = "Working"` is answered almost only by employed people (8,952 employed versus 196 unemployed in the labour force).
- **Identifiers and design columns:** `person_number` and `psu` are not predictors. `household_id` and `person_weight` are retained in the file for the grouped split and survey weighting, but are never used as predictors.
- **Replaced columns:** code columns, label columns and raw columns that an engineered feature now represents (for example `age` is replaced by `age_centred`).
- **Not applicable to the population:** columns recorded only for people who already attend an educational institution.

5.3 Train/test split

Problem: people in the same household share every household-level predictor. A row-level random split would put members of the same household on both sides, so the model would be tested on near-duplicates of households it had already seen, overstating performance. (Frick et al., 2026)

Treatment: an 80/20 split of **distinct households**, with every member following their household's assignment, using a fixed seed (67) for reproducibility. (Frick et al., 2026)

Dataset	Households in (train)	Households in (test)	Total Rows in (train)	Total Rows in (test)
model_data	13,406	3,352	23,041	5,701
income_data	10,460	2,615	14,371	3,646

Verification: no household appears on both sides of either split, the training and test rows together equal the full dataset, and the class balance stays close to the full dataset's 64.9% employed in both parts (64.9% training, 64.7% test).

5.4 Preventing preprocessing leakage

Every transformation applied before the split uses a fixed rule or constant, dictionary code mappings, the 15–64 age range, the R100 salary threshold, centring age on 40, and grouping ratios. None of them is estimated from the data as a whole, so applying them before or after the split produces identical values and no information passes from the test rows into training. (Frick et al., 2026)

No scaling or one-hot encoding is stored in the files. Logistic regression and tree-based methods do not require scaling, and R creates dummy columns internally from the factor levels, leaving out the chosen reference level. Any estimating that needs to happen will happen when models are created based on the needs of each model selected.

6. Reproducibility and Deliverables

6.1 The pipeline

The preparation runs in two Quarto documents, in this order. Both read from the files on disk rather than from objects left in memory, so either can be rerun on its own.

File	What it does
Scripts/Data Preprocessing.qmd	Sections 1–9 are for import, cleaning, column selection, special values, record selection, recoding, outlier review, saving the cleaned files and integration. Section 10 is for feature engineering, transformation, encoding and saving the analytical datasets.
Scripts/Train-Test Split.qmd	Target definition, class balance and the household-grouped split are performed here as well saving the training and test files for modelling.

6.2 Datasets produced

File	Rows	Columns	Purpose
Cleaned/D01_clean.csv	45,413	15	Cleaned demographics
Cleaned/D02_clean.csv	45,413	10	Cleaned education variables
Cleaned/D03_clean.csv	45,413	7	Cleaned employment status and salary
Cleaned/D08_clean.csv	20,940	8	Cleaned household connectivity (all households)
Analytical/analytical_dataset.csv	45,413	59	The four datasets integrated, with all engineered features used for Power BI and descriptions
Analytical/model_data.csv	28,742	24	Employment modelling dataset
Analytical/income_data.csv	18,017	13	Income dataset
Analytical/model_train.csv / model_test.csv	23,041 / 5,701	24	Household-grouped 80/20 split of model_data
Analytical/income_train.csv / income_test.csv	14,371 / 3,646	13	Household-grouped 80/20 split of income_data

7. Ethics and Responsible AI

7.1 Privacy and disclosure

The data contain no names or addresses, but people could still be identified if results are reported for very small groups. For example, a table showing one or two people in a small area could point to a real household. For this reason `metro_code`, which splits the provinces into 17 municipalities, was removed, it is more geographic detail than the project needs. Household and person numbers are kept only to join the datasets and to keep each household together when the data is split for training and testing. They are never used to predict anything and say nothing about the people themselves.

7.2 Fairness and sensitive attributes

Population group and sex are kept in the model as controls. Leaving them out would hide inequalities that matter in South Africa, and those inequalities are part of what the project is trying to understand. Both are sensitive, so they need care. In Milestone 3 the model should be tested separately for each group, to check that it works equally well for everyone. Any recommendation must also describe differences as patterns between groups, not as traits of individuals.

7.3 Representativeness and honest reporting

The survey did not give every person an equal chance of being selected, so each record carries a weight (`person_weight`) showing how many South Africans that person represents. The weights are used whenever results describe the population, and they are never changed. The data is a single snapshot of 2024, so the results can only show that things occur together, not that one causes the other. The relationship may even occur in the opposite way, having a job or income makes it easier to pay for internet access. Household wealth is also not accounted for, because the household resources dataset (D09) was not included. Wealth could be the real reason behind some of the patterns we see. (Statistics South Africa, 2025c)

7.4 Responsible AI usage log

Where used	What it was used for	How it was verified
R Code	Helped with drafting code for feature engineering, transformation and encoding, and suggesting candidate features	The whole pipeline was rerun end to end every step has R output showing row counts, missing values and category counts, which the team reviewed
Documentation	Helped rewrite sections to make it sound more professional.	All text was read again and confirmed to be correct.

8. Limitations and Open Decisions

Choice or Limitation	Status
Household wealth is not controlled for (D09 excluded)	Recorded as a limitation. D09 is household-level and joins the same way as D08, so an asset-based wealth control can be added in as a robustness check without changing the grain
The employment outcome comes from a single source (D03)	D11 cannot be linked since the IDs do not match, so no second source is available
Digital access is measured per household, not per person	Findings refer to living in a connected household, not to personal internet use
matric_plus is missing for 497 people in the modelling dataset	Their education is “Other/unknown”. education_band and edu_digital_group keep them as their own level a model using matric_plus drops them
Area A (digital access and education) datasets	The prepared modelling datasets cover Areas B, C and D. The Area A subsets (adults 25–64, youth 15–24) are defined and can be filtered from analytical_dataset.csv

9. References

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